Chinnamuthu S

Curriculum Vitae

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Education

Dec 2019-April 2025 **PhD, Mathematics**, *National Institute of Technology Goa*, Goa, India, CGPA: 9.25.

Thesis Title: Efficient Computational Algorithms for Randomized Neural Networks.

Advisor: Dr. Ravi Prasad K.J, Associate Professor, Department of Applied Sciences.

Specialization: Machine Learning

2012–2014 **MSc, Mathematics**, *Government Arts College, Coimbatore (Affiliated to Bharathiar University)*, Coimbatore, Tamilnadu, India, CGPA: 6.79.

2009–2012 **BSc, Mathematics**, *Government Arts College, Coimbatore (Affiliated to Bharathiar University)*, Coimbatore, Tamilnadu, India, CGPA: 7.40.

2007–2009 **HSC**, *Government Higher Secondary School Begarahalli (State Board)*, Dharmapuri, Tamilnadu, India, Percentage marks: 72.3%.

2007 **SSLC**, *Government Higher Secondary School Begarahalli (State Board)*, Dharmapuri, Tamilnadu, India, Percentage marks: 73.2%.

Academic Achievements & Awards

2022–2024 MHRD-Senior Research Fellowship (SRF), National Institute of Technology Goa.

2020–2021 MHRD-Junior Research Fellowship (JRF), National Institute of Technology Goa.

2019 Qualified GATE (Mathematics).

Dec 2018 Qualified CSIR-UGC NET (LS)(Mathematical Sciences).

June 2016 Qualified CSIR-UGC NET (JRF)(Mathematical Sciences), Rank: 145.

Research Experience

June 2026-Till date Post Doctoral Researcher, Centre for Responsible AI (CeRAI), Wadhwani School of Data Science and AI (WSAI), Indian Institute of Technology (IIT) Madras, Chennai, Tamilnadu, India.

Mentor: Dr. B. Ravindran, Head, CeRAI, WSAI, and DSAI.

Research Focus: Interpretable and Explainable AI.

May 2025-June 2025 Research Intern, Numerical Algorithms and Tensor Learning (NATL) Lab, Indian Institute of Science (IISc), Bangaluru, Karnataka, India.

Mentor: Dr. Ratikanta Behera, Assistant Professor, Department of Computational and Data Sciences (CDS).

Dec 2019-April 2025 Doctoral Researcher, National Institute of Technology Goa, India.

Research Focus: Developing efficient algorithms for randomized neural networks by incorporating numerical linear algebra, optimization, and probabilistic techniques.

Key Contributions:
- Designed scalable and computationally efficient algorithms for training deep stochastic configuration networks.
- Improved performance of extreme learning machines using filtering techniques and iterative methods.
- Collaborated with multidisciplinary teams on machine learning applications in fluid dynamics.

Advisor: Dr. Ravi Prasad K.J, Associate Professor, Department of Applied Sciences

Work Experience

July 2025 – May 2026: Assistant Professor: Amrita Vishwa Vidyapeetham, Amaravati Campus, Kuragallu Village, Mangalagiri (M), Guntur, Andhra Pradesh – 522503, India.

Dec 2018 – Dec 2019: Guest Lecturer: Puratchi Thalaivi Amma Government Arts and Science College, Palladam -641664, Tamilnadu.

Sep 2018 – Dec 2018: Guest Lecturer: Government Arts College, Coimbatore -641018, Tamilnadu.

Teaching Assistantship During PhD

Courses: Assisted in course delivery for the below-provided courses.
- Linear Algebra
- Probability and Statistics
- Complex Analysis
- Optimization Techniques

Publications

Journal Articles

2025 **Subramani, C.,** Jagannath, R. P. K., & Kuppili, V., (2025). Randomized Gauss–Seidel iterative algorithms for Extreme Learning Machines. *Physica A: Statistical Mechanics and its Applications*, 130515, (**Q2, Impact factor 2.8**).

2025 **Subramani, C.,** Jagannath, R. P. K., & Kuppili, V., (2025). Improving Deep Random Vector Functional Link Networks through computational optimization of regularization parameters. *Engineering Applications of Artificial Intelligence*, 148, 110389., (**Q1, Impact factor 7.5**).

2024 **Subramani, C.** Jagannath, R. P. K., & Kuppili, V , Enhancing deep stochastic configuration networks: Efficient training via low-rank matrix approximation, *Information Sciences*, 690, 121519., (**Q1, Impact factor 8.1**).

2023 Naik, S. M.**, Subramani, C.,** Jagannath, R. P. K., & Paul, A, Exponential filtering technique for Euclidean norm-regularized extreme learning machines, *Pattern Analysis and Applications*, 26(3), 1453-1462, (**Q2, Impact factor 3.7**).

2023 Singh, M., Ragoju, R., Shiva Kumar Reddy, G., **& Subramani, C.** , Predicting the effect of inertia, rotation, and magnetic field on the onset of convection in a bidispersive porous medium using machine learning techniques, *Physics of Fluids*, 35(3) (**Q1, Impact factor 4.6**).

Communicated Journal Article

2025 **Chinnamuthu Subramani,** Shraddha M. Naik, Ravi Prasad K. Jagannath, and Venkatanareshbabu Kuppili, Efficient Regularization in Extreme Learning Machines Using the Gauss-Seidel iterative method, *Physica A: Statistical Mechanics and its Applications*.

In Conference Proceedings

2023 **Subramani, C.,** Kuppili, V., Keshavamurthy, B. N., Jagannath, R. P. K., & Prashanth, G. R., The Least Square QR Method Improves Extreme Learning Machine, In 2023 IEEE World Conference on Applied Intelligence and Computing (AIC) (pp. 1-6). IEEE.

Workshops & Presentations

Workshops

11-23, July,2022 **Advanced Topics in Statistical Learning**, Indian Statistical Institute (ISI), Kolkata

24- 28 April 2023 **Medical Image Analysis Using Artificial Intelligence**, Gayatri Vidya Parishad College of Engineering, Visakhapatnam

Presentations

29-30, July,2023 **The Least Square QR Method Improves Extreme Learning Machine**, *In 2023 IEEE World Conference on Applied Intelligence and Computing (AIC)* Ashoka Institute of Technology & Management, Varanasi, India

Skills

Programming: Python, MATLAB

Technical Writing: LaTeX, Academic Research

Machine Learning Tools: Scikit-learn, TensorFlow, PyTorch

Referees

Dr. Ravi Prasad K.J
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Dr. Venkatanareshbabu K
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